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9. STUDY OF RELATION BETWEEN FREQUENCY AND LENGTH OF A GIVEN WIRE UNDER CONSTANT TENSION USING SONOMETER

AIM

To study the relation between frequency and length of a given wire under constant tension using a sonometer.

APPARATUS REQUIRED

Sonometer, six tuning forks of known frequencies, Metre scale, rubber pad, paper rider, hanger with half – kilogram masses, wooden bridges

FORMULA

The frequency n of the fundamental mode of vibration of a string

is given by $n = \frac{1}{2l}\sqrt{\frac{T}{m}}\ Hz$

a) For a given m and fixed T.

$n \propto \frac{1}{l}$ (or) nl = constant

where $n \rightarrow$ Frequency of the fundamental mode of vibration of the string (Hz)

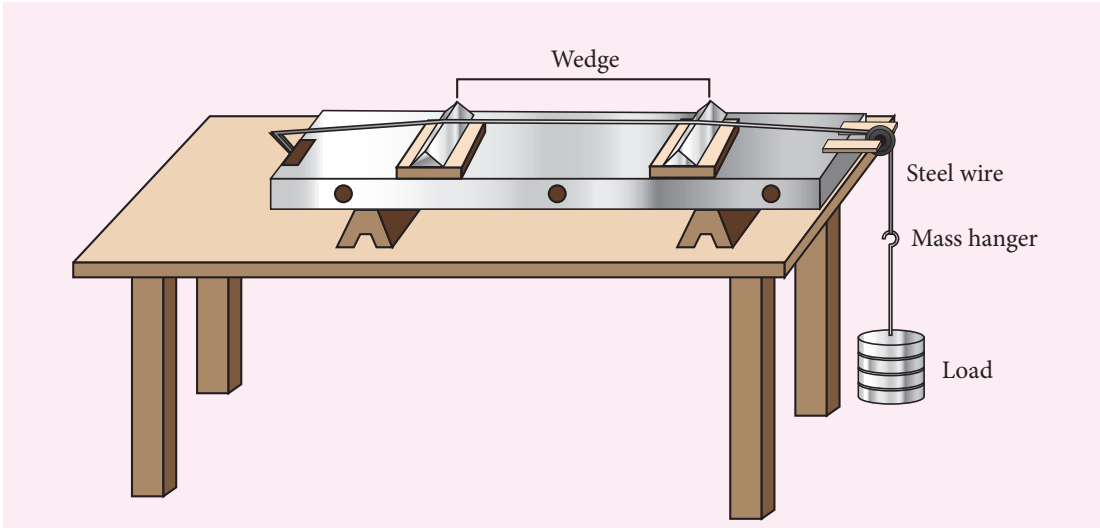
$m \rightarrow$ Mass per unit length of the string ($kg\ m^{-1}$)

$l \rightarrow$ Length of the string between the wedges (m)

$T \rightarrow$ Tension in the string (including the mass of the hanger) = Mg (N)

$M \rightarrow$ Mass suspended, including the mass of the hanger (Kg)

DIAGRAM



SONOMETER - STUDY OF RELATION BETWEEN FREQUENCY AND LENGTH OF A GIVEN WIRE UNDER CONSTANT TENSION USING SONOMETER